



Anesthetic complications

It is any deviation from the normally expected pattern during or following the administration of LA (The normal reaction should be lack of pain sensation without any adverse side effects).

These complications vary in intensity from mild to severe. In a mild complication there is slight change from normally expected pattern that may reverse itself without any specific TTT. While a severe complications manifests itself by a pronounced deviation from normally expected pattern and requires a definite TTT.

Local complications of LA

1- Facial nerve paralysis:

Transient facial nerve paralysis is commonly caused by introduction of LA solution into the capsule of the parotid gland which is located at the posterior border of the ramus of the mandible during inferior alveolar nerve block or Akinosi tech

When it happens the patient will loss the motor function of the muscles of facial expression.

Loss of ability to close the eye and inability to bow the cheek or control the lip function.

TRANSIENT FACIAL PALSY

- Encountered during IANB
- Due to injection of LA into parotid gland if needle injected too backwards
- Temporary paralysis of facial nerve
- Effect wears off over a period of time(<3Hrs) during which eye needs to be protected(eye patch)



Temporary facial nerve paralysis

Management :

- Assurance of the patient
- Protect the eye from dryness

2- Muscle trismus :

This may result from injection into muscles or ligaments or from needle borne infection or contaminated solution.

TTT depends on the cause + slight exercise and pain killer /muscle relaxant in rare cases

3- Pain :

May be due to one of the following :

- A- Using non-isotonic sol.
- B- Very cold sol.
- C- Too rapid injection
- D- contaminated sol.
- E- Multiple needle puncture
- F- Injection into a muscle or ligament

4- Infection

Caused by non-sterile needle or sol. Or from carrying bacteria from non-sterile MM into the deeper tissues with the tip of the needle.

5- Hematoma and Ecchymosis

Hematoma is the effusion of blood into the tissues as a result of punctured vessel.

It is usually associated with Zygomatic or infra-orbital injection.

As a result of breakdown of HB in the extravasated blood, discoloration of the skin occur.



TTT consists of cold application and applying pressure if possible and noticed at beginning of bleeding

Few days or week after that it disappears gradually

6- Sloughing and ulceration

Traumatic ulcers may appear on the palate as a result of large volume of LA sol injected .

Chewing the lip following inferior alveolar nerve block especially in children or pinching of the lip between the forceps and the teeth.





7- Prolonged anaesthesia: Caused by

- A) Contamination of the anesthetic solution by alcohol or other germicidal solution
- B) Needle trauma to the nerve and this is manifested at the time of injection by sharp hot pain felt by the pt.

Management : Assurance of the pt that it will resolve without any ttt.

8- Broken needle : The frequency of this complication has been greatly reduced by introduction of disposable needles.

However these should be avoided to prevent this complication:

- A) Do not attempt to force a needle against resistance.
- B) Do not use needle of too fine gauge.

C) Do not attempt to change the direction of the needle while it is embedded in tissues. Always withdraw the needle and redirect it.

D) Do not insert the needle fully inside the tissues.

E) Do not surprise the pt with sudden unexpected needle insertion.

Management :

- Remain calm and ask the pt to keep the mouth open and if there is protruded part of the needle, remove with hemostat.
- If no part protruded it will be removed surgically.

